A1. Equivalence Class Partitioning

Equivalence Class Partitioning (ECP) divides input data into groups (classes) where the system is expected to behave similarly.

We divide each input into valid and invalid classes and select representative values from each class.

1. Credit hours

According to FR-01 and FR-02:

- Current credits: 0–18 (inclusive).
- New course credits: 1–4 (inclusive).
- Combined credits must not exceed 18.

Class ID	Equivalence class	Representative input	Expected output
EC01	Valid current credits	10	Accepted
EC02	Invalid current credits (below range)	-1	ValueError
EC03	Invalid current credits (above range)	19	ValueError
EC04	Valid course credits	3	Accepted
EC05	Invalid course credits (below range)	0	ValueError
EC06	Invalid course credits (above range)	5	ValueError

EC07	Valid combined credits	15 + 3	True
EC08	Invalid combined credits	16 + 3	False

2. Outstanding dues

According to FR-04 and FR-15:

Class ID	Equivalence class	Representative input	Expected output
EC09	No outstanding dues	Rs. 0	Registration allowed if other conditions pass
EC10	Positive dues	Rs. 1	DUES_BLOCK
EC11	Negative dues	Rs. -1	ValueError

3. Prerequisites

According to FR-03 and FR-06:

Assume required prerequisites are `{CS2001, MATH101}`.

Class ID	Equivalence class	Representative input	Expected output
EC12	No prerequisites required	Empty set	True

EC13	All prerequisites completed	`{CS2001, MATH101}`	True
EC14	Some prerequisites completed	`{CS2001}`	False
EC15	No prerequisites completed	`{}`	False

4. Course availability

According to FR-08:

Assume course capacity is 30.

Class ID	Equivalence class	Representative input	Expected output
EC16	Seats available	Enrolled = 29	REGISTERED
EC17	Course full	Enrolled = 30	WAITLISTED
EC18	Invalid enrollment	Enrolled = 31	ValueError
EC19	Invalid capacity	Capacity = 0	ValueError